Can Consistent Forces Recover Molecular Shapes?

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Abstract

Learned molecular forces describe pushes and pulls on atoms and can accelerate searches for stable molecular shapes, but accurate force predictions do not guarantee recovery of the correct shape. Three graph neural network formulations were evaluated on aspirin: forces calculated from negative energy slopes, forces predicted directly, and direct forces trained to agree with energy-derived forces. Consistency strength 0.1 achieved a validation force error of 38.024 meV/Å versus 38.186 for direct prediction, while requiring 1.985 times its CPU fitting-and-scoring time. All five tested model states, including the energy baseline, recovered zero of ten targets, although every target shape appeared in training. All 23 endpoints meeting the force stopping rule for direct and consistency models had atom groups separated beyond the model's 5 Å interaction range. Zero recovery prevents a ranking of methods; the diagnostics identify model and procedure limitations, while controlled movement-cap tests are needed to assess the effect of excessive atom movements.

1 Introduction

Repeated calculations of molecular energies and forces are expensive, so learned approximations can reduce simulation cost. Training changes a model's adjustable numbers, called parameters, to match reference examples.

Energy is a number assigned to an atomic arrangement. A stretched spring gives a useful picture: stretching it further stores more energy, and its pull points toward lower energy. Energy slope means how quickly energy changes when an atom moves; the energy baseline calculates forces from the negative slopes. The direct model predicts force arrows themselves, while consistency training penalizes disagreement with forces calculated from negative energy slopes. Direct prediction skips the slope calculation and can be cheaper, but its forces need not follow any single energy.

The objective is to compare correct-shape recovery at comparable force error, with computing cost measured separately. Recovery requires the search to reach its assigned target shape and meet the force stopping rule; neither condition alone establishes success.

2 Literature Review

Landscape17 supplies molecular shapes, reference forces and paths between stable arrangements [1]. It motivates checking final shapes beyond prediction errors, while Bigi et al. describe problems when direct forces do not follow one energy function [2]. Together, these studies motivate separating a force-based stop from structural success.

The models are built with E3x, software for networks that obey three-dimensional symmetry rules [3]. MACE supplies a separate established energy-derived model for predicting energies and forces [4]. Earlier work already studies physical constraints in training losses [5]. The unresolved

question here is whether consistency improves recovery at comparable force error; consistency training itself is not new.

3 Data and Preprocessing

The dataset contains aspirin from Landscape17 [1], with 21 atoms in each molecule. Reference energies and forces come from hybrid density-functional theory, a detailed calculation based on electron physics, so these reference labels are calculated answers rather than exact experimental truth.

A transition state is a high-energy arrangement with two downhill branches toward stable arrangements, called minima. A connection group contains both branches' records, which stay in the same partition to avoid separating the two sides of one transition state. Training adjusts parameters, validation selects settings and saved models, and the test partition is reserved for later trained-model evaluation.

The training set contains 26 connections and 4,891 records. Validation contains five connections and 974 records; testing contains six connections and 1,127 records. Training-only normalization puts energy and force errors on comparable scales, while shared minima mean that records are not independent examples of new destinations. Each branch targets the minimum it should reach; empirical classifications corrected 26 of 74 targets previously assigned from filename ordering. Earlier model-free software checks parsed test coordinates and labels but did not choose models or solver settings; no trained model has been scored on the test partition.

4 Methodology

4.1 Model Architecture

The E3x models are graph neural networks in which atoms are nodes and nearby atom pairs form links. Successive calculation stages, called layers, pass learned information along these links. Inputs are atom types and three-dimensional positions; outputs are one molecular energy and a force arrow for each atom. The three compared model types use 64 feature channels containing learned numbers and vectors, two message-passing rounds and eight radial functions describing pair distances.

Links extend to 5 Å, where an angstrom (Å) is 10^{-10} metres. Internal features carry scalar numbers and vectors with direction. Rotating a molecule rotates its predicted forces without changing its energy, properties called equivariance for forces and invariance for energy. The energy baseline has 43,521 parameters; models with separate energy and force outputs have 43,649.

The energy baseline calculates $\widehat{\mathbf{F}}^E = -\nabla_{\mathbf{R}}\widehat{E}$. Here, $\widehat{E}$ is predicted energy, $\mathbf{R}$ contains atom positions, and the gradient gives energy slopes along position coordinates. Both the direct and consistency models also predict energy trained against reference labels, so they are not force-only models. Ordinary direct training has no agreement penalty, while consistency training adds one. In this architecture, an isolated atom has no directional message, so the direct-force output is zero.

4.2 Training and Evaluation

The loss combines normalized squared energy error, normalized squared force error and, where applicable, force–energy disagreement; training adjusts parameters to reduce this number:

$$\mathcal{L} = w_E \,\mathrm{mean}(e_E^2) + w_F \,\mathrm{mean}(e_F^2) + \lambda \,\mathrm{mean}\left[\left((\widehat{\mathbf{F}}^D - \widehat{\mathbf{F}}^E)/s_F\right)^2\right]. \tag{1}$$

Here, e_E and e_F are energy and force errors divided by training-derived scales; superscript D means direct prediction. The scales are $s_E \approx 0.21584$ eV per molecule and $s_F \approx 0.126886$ eV/Å. Supervised weights are $w_E \approx 0.01458$ and $w_F \approx 1.985$; exact values remain in the configurations. The energy baseline omits the final term; direct training uses $\lambda = 0$, while consistency training uses positive strength λ.

Final E3x fits use Adam, an optimization method that updates parameters, with learning rate 0.003 and 64-bit floating-point arithmetic. Each update samples four records that can repeat, giving

800,000 sampled exposures across 200,000 updates. Two seeds specify random initialization and sampling, providing a limited repeatability check. E3x fits used an arm64 macOS CPU, Python 3.12.13, JAX/JAXlib 0.9.2 and E3x 1.0.2.

Validation occurs every 500 updates, including zero, and BEST is the first saved model state, or checkpoint, with minimum validation force error.

For each validation connection, force mean absolute error (MAE) averages the absolute prediction error across every record, every atom and all three coordinate directions. The reported score averages these five connection scores equally; smaller values are better.

Scores use meV/Å: a millielectronvolt is one thousandth of an electronvolt, an energy unit, and energy divided by distance measures force.

Recovery additionally requires maximum absolute predicted force component ≤ 0.01 eV/Å and aligned structural RMSD ≤ 0.05 Å. Root-mean-square deviation (RMSD) measures typical atom-position disagreement after rotation, translation and allowed atom-symmetry matching. Small predicted force is a stopping condition; it is not small prediction error.

An informative force fit requires validation MAE ≤ 50 meV/Å, approximately 20% below zero-force prediction. The recovery adequacy criterion requires the selected energy baseline to recover at least one branch and exceed the measured unchanged-start count. This checks whether the procedure improves the starting structures.

4.3 Relaxation Procedure

Adaptive force-following relaxation proposes $\mathbf{R}_{\mathrm{trial}} = \mathbf{R} + \eta\widehat{\mathbf{F}}$, where η converts forces into coordinate changes. It starts at 0.2 Å^2/eV and can grow to 1.0. Backtracking halves a failed trial's scale; each branch permits 1,500 force attempts.

A trial is accepted immediately if its force meets the stopping condition. Otherwise, acceptance requires $\sum_{i,\alpha} \widehat{F}_{i\alpha}^{\mathrm{trial}} \widehat{F}_{i\alpha}^{\mathrm{current}} \geq -10^{-12}$ (eV/Å)^2, summing over atoms i and coordinate components α. There is no energy-decrease test or per-atom movement cap, and no FIRE or BFGS optimizer comparison was performed.

5 Results and Discussion

5.1 Baseline Development

Early E3x training on a 128-record subset plateaued, but MACE, which also derives forces from energy, fitted those records to 0.001515 of the zero-force squared error. Different settings prevent a controlled comparison, but the plateau did not establish an unavoidable data or energy-derived-force limit.

Later 1,000-update energy and direct fits selected update zero as BEST: training improved energy but worsened force predictions.

Three 200,000-update energy baselines were then completed (Figure 1). A used width 32 and learning rate 0.001; B used width 64 and 0.003. Both used equal normalized energy/force weights, while C retained B's structure and rate but used the balanced weights in Section 4.2. Width counts internal feature channels.

C reached 34.587 meV/Å, versus 40.370 for B and 42.108 for A, and passed the 50 meV/Å adequacy rule. Predicting zero force gives 62.504 meV/Å, so C reduces that error by 44.664%. However, C's energy error is higher than B's: 126.990 versus 28.178 meV per molecule. C was selected for force accuracy despite this energy trade-off; its recipe was transferred without retuning the other model types.

5.2 Predictive Accuracy and Computational Cost

Reusing C seed zero and completing nine additional fits provided two seeds per row in Table 1. Consistency strength 0.1 reduces normalized force–energy disagreement squared error to 0.49–0.61,

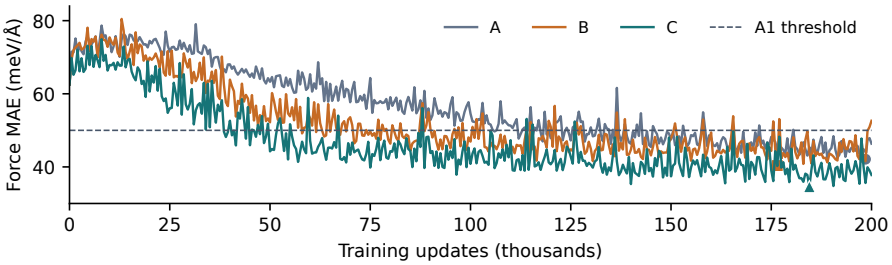


Figure 1: Energy-baseline validation, seed 0. The triangle marks C's BEST at update 184,500; the dashed line is the declared 50 meV/Å adequacy rule. Each point uses all 974 validation records, with equal connection weighting.

versus 116–379 for direct training. Its measured fit-plus-score time on CPU is 1.985 times direct prediction, excluding other diagnostics and overhead. This is not inference speed. Direct training also yields lower energy MAE than the energy baseline, 76.684 versus 128.725 meV per molecule; the cause was not isolated.

Table 1: Validation error at BEST. Force columns use meV/Å; energy uses meV per molecule. Seed limits show two observed values, not confidence bounds.

Model	Mean force MAE	Seed min–max	Mean energy MAE
Energy-derived	34.893	34.587–35.198	128.725
Direct	38.186	37.291–39.082	76.684
Consistency $\lambda = 0.1$	38.024	36.773–39.275	87.006
Consistency $\lambda = 1$	34.683	33.723–35.643	112.341
Consistency $\lambda = 10$	39.725	39.060–40.390	161.662

Only strength 0.1 meets the declared matching rule against direct prediction: the mean-error gap must be within 10% of the smaller mean and within both seed spans. A span is maximum minus minimum.

BEST is optimistic because it selects among 401 checks. Over updates 150,000–200,000, the last 50,000 updates, baseline errors vary by 2.2–3.1 meV/Å standard deviation across correlated checks. Nine subsequent fits finish 0.1–7.4 meV/Å above BEST. The 0.162 meV/Å gap meets the matching rule but is smaller than the observed variation, so matching does not establish equivalence.

MACE Run A independently scores 5.349 meV/Å with equal connection weighting after 2,048 epochs on the same 4,891 training records. Its 10,014,720 training exposures are 12.52 times E3x's 800,000 sampled exposures. Sampling, architecture, optimization and hardware differ; exposure alone does not explain the accuracy gap. This is an external reference, not a controlled efficiency comparison; MACE recovery remains unmeasured.

5.3 Structural Recovery

A relaxation procedure repeatedly moves atoms along predicted forces. Five saved E3x checkpoints were tested from the same ten branch starts across five validation connections. Each branch has its own target, and all six distinct target minima appear in training. Leaving the starts unchanged recovers zero targets, while every evaluated checkpoint also recovers 0/10 (Table 2). The selected energy baseline therefore fails the recovery adequacy criterion, and successful recovery cannot rank the methods.

Across 50 model/branch outcomes, 25 stop with wrong geometry and 25 fail to converge. Reusing the same five connections does not create 50 independent paths. The reference-start controls favour the energy checkpoint under this procedure, but do not isolate a consistency-penalty effect.

Table 2: All checkpoints recover 0/10 original targets. "Wrong stop" meets the predicted-force rule but misses the target shape. Reference-start controls begin at the six target minima; they are separate follow-up checks, called post-hoc analyses.

Checkpoint	Wrong stop	No convergence	Reference-start recovery
Energy C, seed 0	2	8	4/6
Direct, seed 0	4	6	1/6
Direct, seed 1	4	6	0/6
Consistency 0.1, seed 0	6	4	0/6
Consistency 0.1, seed 1	9	1	0/6

5.4 Endpoint Diagnostics

Every model exceeds the stopping force threshold at all 11 stored aspirin minima, which include the six targets; reference labels stay below 2.745 meV/Å. The 155 evaluations cover five checkpoints, each evaluated at ten starts, ten endpoints and eleven minima; all 50 stopping flags were reproduced. This mismatch at exact reference shapes does not exclude nearby shapes meeting both tolerances.

All 23 force-converged direct/consistency endpoints have disconnected 5 Å neighbour graphs, with atoms in different fragments beyond the model's interaction cutoff. These are not intact molecular shapes, although internal fragment forces can remain. Only seven endpoints have every atom isolated, where the direct-force output is exactly zero automatically; this does not establish a correct molecular minimum.

Fixed post-hoc traces replayed seed-0 direct, consistency-0.1 and energy models on 18/plus, plus energy on 13/plus. These four diagnostic cases reproduced saved endpoints; they are not a representative sample. Two separated, with largest accepted atom movements of approximately 437 Å for direct prediction and 268 Å for energy-derived forces. The other two remained connected, with maxima approximately 0.086 Å for consistency and 0.0061 Å for energy-derived forces. The separated energy trace ended without convergence; both force-converged energy endpoints remained connected.

5.5 Representation Study

Feedback from a faculty member, a professor of materials science, suggested comparing models with and without rotation symmetry; the internal-feature check is related but different. E64 is the existing width-64 model with scalar numbers and directional vectors; its two fits were reused. Four new scalar-only fits match its width (S64) or approximate parameter count (SP). All models retain force rotation symmetry, so this is not a no-symmetry comparison. E64 had lower error in both seeds; mean BEST force MAEs were 34.893, 39.542 and 41.358 meV/Å for E64, S64 and SP, respectively. E64-selected settings were transferred without scalar retuning; these preliminary results await independent interpretation review and establish neither feature causality nor better recovery.

6 Conclusion and Future Work

Force-error matching did not deliver recovery under this procedure, and zero recovery cannot establish method equivalence.

Limitations. Evidence covers aspirin, five validation connections and two seeds, with shared minima. Validation selects settings and BEST; test generalization is unestablished. These checks do not isolate the procedure's contribution or establish real molecular break-up. Failed and incomplete trials remain recorded.

Changes from the proposal. Two seeds were completed rather than the proposed three, and paracetamol replication is deferred until aspirin recovery is feasible. A revised adequacy protocol followed the preserved small-subset failures; responsibilities changed, and the representation check adds an extension. The models use a 5 Å cutoff; faculty suggested studying interactions beyond 7 Å as future work. These results do not show that missing long-range interactions caused the failures.

Planned Work. First, verify the MACE loader against saved predictions and run the unchanged ten-start and six-reference controls. Second, use these controls to assess a separately declared per-atom movement cap on the same checkpoints and starts, labelling reused-validation analysis as post-hoc.

Third, measure closed-loop work and prediction time on identical hardware, declaring any matched-wall-time training budget before new runs. Work adds products of matching force and displacement components around a closed path; forces from one smooth energy have an exact zero closed-loop integral. Numerical integration can introduce residual error.

Team contributions and reproducibility. Shedrack completed data preparation, model/training integration and current MACE/E3x experiments. Olufunke authored the recovery evaluator. Emmanuel Aaron leads physical/cost analysis; Muhammed Idris leads MACE loading, prediction checks and independent reproduction.

Research code and evidence: GitHub,
`github.com/Shedrackeze002/idl-molecular-force-consistency`, commit 3fa5969.

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